



ADVANCE ANALYTICS

Part A1: Sentiment Analysis of City Pass Reviews

1. Introduction

Determining customer satisfaction and addressing sentiment analysis of customer reviews is crucial to boost loyalty and experience. From the City Pass platform, we have an extensive feedback array for attractions such as museums, zoos, and cruises. We need to develop the best out of four classifiers based on NLP: **Logistic Regression**, **Multinomial Naive Bayes**, **CNN** and **BERT** model-with the task of distinguishing positive and negative sentiment.

2. Dataset and Preprocessing

The dataset contains reviews from six City Pass attractions with labelled binary sentiment (1 = positive, 0 = negative). We merged these datasets into one corpus.

Several important preprocessing steps were taken in terms of assuring the data quality and the model effectiveness:

- Adjusting the text to lowercase to remove variance in a dependence with case.
- Removing punctuation and numerical tokens to decrease noise and provide relevant language.
- Using NLTK to tokenize and remove stop words to further strengthen focus on relevant material.
- Creating a cleaned text feature (cleaned_text) useful for improved accuracy in sentiment analysis.

TF-IDF vectorisation (top 5,000 tokens) was used for classical models such as Logistic Regression, and Naive Bayes. CNN used tokenised sequences, while BERT utilized pre-trained transformer embeddings (bert-base-uncased).

3. Models Implemented

3.1 Logistic Regression

- TF-IDF feature vectors.

- Baseline model; easy to understand and interpret results.
- Very short training/inference times, well-suited for lightweight situations.
- Disadvantage of not understanding sensitive contexts, especially sentiment shifts that are more nuanced.

Logistic Regression Model Performance:				
Accuracy: 0.9752650176678446				
Classification Report:				
	precision	recall	f1-score	support
Negative	0.99	0.95	0.97	128
Positive	0.96	0.99	0.98	155
accuracy			0.98	283
macro avg	0.98	0.97	0.97	283
weighted avg	0.98	0.98	0.98	283

Fig 2. Confusion Matrix of Logistic Regression Predictions

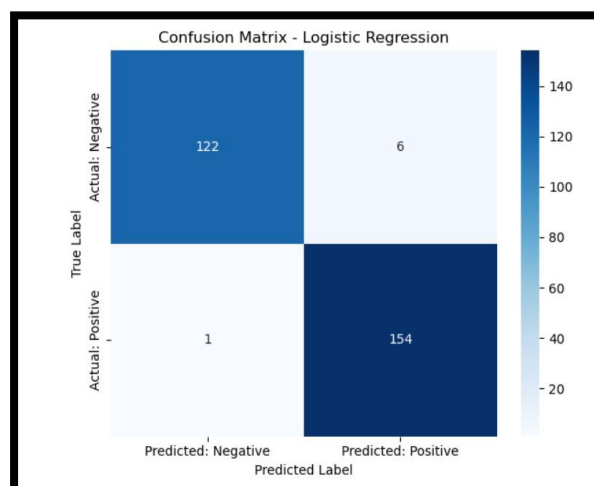


Fig 2. Confusion Matrix of Logistic Regression Predictions

3.2. Multinomial Naive Bayes

- Also used TF-IDF feature vectors.
- Assuming that features are independent, it will limit performance in an NLP context.
- Inference time is quick, great for making real time predictions.

- Struggles with negation and reviews that are unable to be read as sarcasm due to assuming words independently.

Naive Bayes Model Performance:				
Accuracy: 0.9823321554770318				
Classification Report:				
	precision	recall	f1-score	support
Negative	1.00	0.96	0.98	128
Positive	0.97	1.00	0.98	155
accuracy			0.98	283
macro avg	0.98	0.98	0.98	283
weighted avg	0.98	0.98	0.98	283

Fig 3. Classification Report for Naive Bayes Model Performance

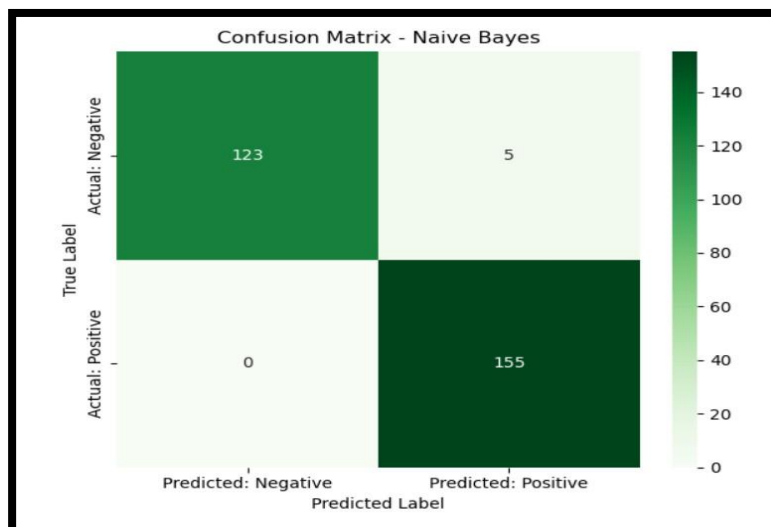


Fig 4. Confusion Matrix of Naive Bayes Predictions

3.3. Convolutional Neural Network (CNN)

- Used embedding layer, 1D convolutions, max pooling.
- Good at recognizing local n patterns (e.g., "not worth it").
- High accuracy, does not require much computing.
- Not as good for reviews, good at capturing short range language patterns.

CNN Model Accuracy: 0.9823321554770318				
Classification Report:				
	precision	recall	f1-score	support
Negative	0.99	0.97	0.98	128
Positive	0.97	0.99	0.98	155
accuracy			0.98	283
macro avg	0.98	0.98	0.98	283
weighted avg	0.98	0.98	0.98	283

Fig 5. Classification Report for CNN Model Performance

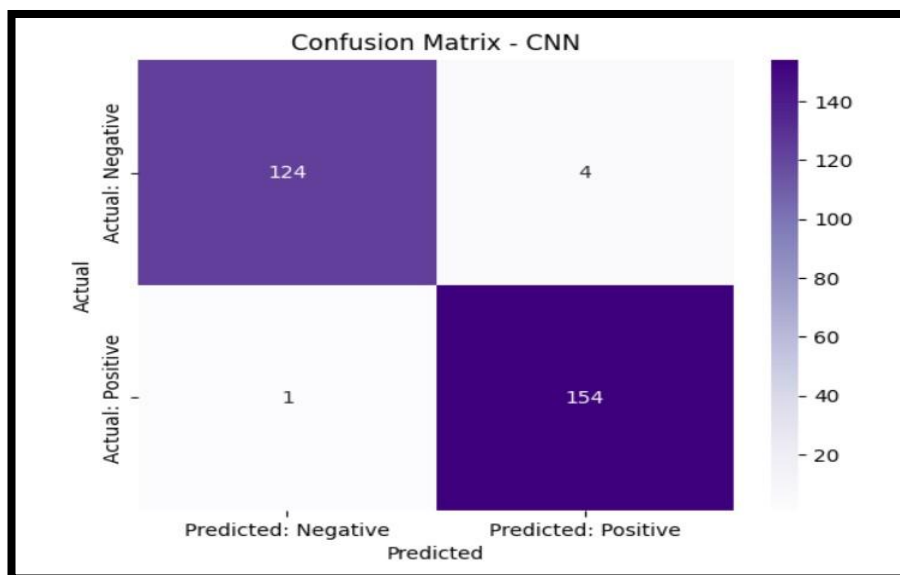


Fig 6. Confusion Matrix of CNN Predictions

3.4 BERT

- Fine-tuned the pre-trained bert-base-uncased model through Hugging Face.
- Used token level self-attention mechanisms and positional embeddings.
- Apply WordPiece tokenization; maximum sequence length of 256.
- Longer to train but was exceptional at capturing nuanced context (i.e., sarcasm and negation).

BERT Test Accuracy: 0.9965				
Classification Report:				
	precision	recall	f1-score	support
Negative	1.00	0.99	1.00	128
Positive	0.99	1.00	1.00	155
accuracy			1.00	283
macro avg	1.00	1.00	1.00	283
weighted avg	1.00	1.00	1.00	283

Fig 7. Classification Report for BERT Model Performance

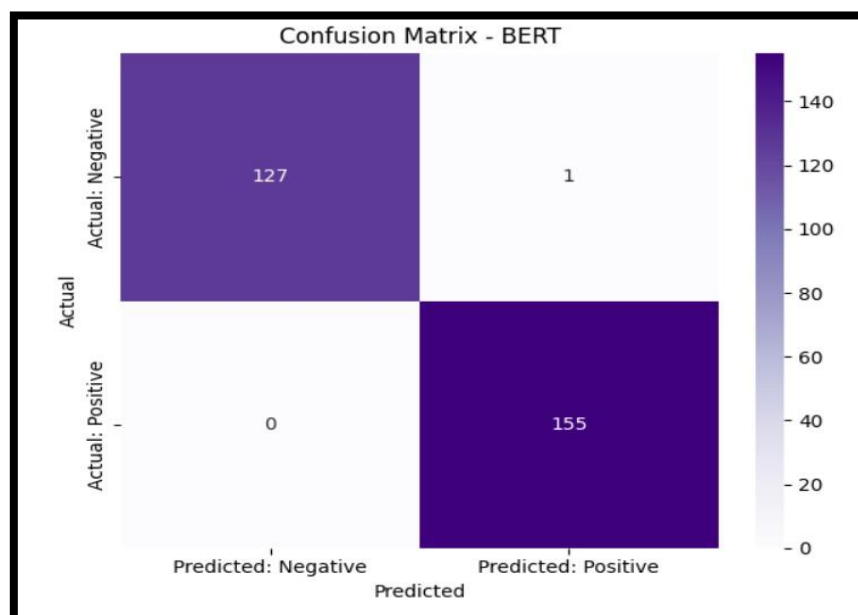


Fig 8. Confusion Matrix of Logistic Regression Predictions

4. Evaluation Metrics

All models were divided into an 80% training - 20% test grouped of data and executed based on a fixed random state based on my student number (36494631) to allow experiment reproducibility.

The evaluation metrics for performance included:

- **Accuracy**
- **Precision, Recall, and F1-Score** (per class)
- **Confusion matrices** for more in-depth error analysis

Metrics were calculated using scikit-learn for the traditional models and deep learning frameworks (TensorFlow) for the CNN and BERT models.

5. Results Summary

Model	Accuracy (%)	F1-Score	Precision	Recall	Notes
Logistic Regression	97.53%	0.978	0.963	0.994	Strong baseline, interpretable
Naive Bayes	98.23%	0.984	0.969	1	Fast predictions, lightweight
CNN	98.23%	0.984	0.975	0.994	Effective phrase recognition
BERT	99.65%	1	1	1	Best at nuanced classification

Fig 9. Summary Table of Model Performance Metrics

Model	Accuracy	Neg class (P / R / F1)	Pos class (P / R / F1)	Errors (FP/FN)	Notes
Logistic Regression	97.53%	0.99 / 0.95 / 0.97	0.96 / 0.99 / 0.98	6 / 1	Strong baseline, interpretable
Naive Bayes	98.23%	1.00 / 0.96 / 0.98	0.97 / 1.00 / 0.98	5 / 0	Fast, lightweight
CNN	98.23%	0.99 / 0.97 / 0.98	0.97 / 0.99 / 0.98	4 / 1	Effective phrase-level patterns
BERT	99.65%	1.00 / 0.99 / 1.00	0.99 / 1.00 / 1.00	1 / 0	Best nuanced understanding

Fig 10. Class-wise Metrics & Error Counts

5.1 Logistic Regression

- Accuracy: 97.53%
- Negative class: P = 0.99, R = 0.95, F1 = 0.97
- Positive class: P = 0.96, R = 0.99, F1 = 0.98
- Errors: 1 FN / 6 FP

5.2 Multinomial Naive Bayes

- Accuracy: 98.23%
- Negative class: P = 1.00, R = 0.96, F1 = 0.98
- Positive class: P = 0.97, R = 1.00, F1 = 0.98
- Errors: 0 FN / 5 FP

5.3 Convolutional Neural Network (CNN)

- Accuracy: 98.23%
- Negative class: P = 0.99, R = 0.97, F1 = 0.98
- Positive class: P = 0.97, R = 0.99, F1 = 0.98
- Errors: 1 FN / 4 FP

5.4 BERT

- Accuracy: 99.65%
- Negative class: P = 1.00, R = 0.99, F1 = 1.00
- Positive class: P = 0.99, R = 1.00, F1 = 1.00
- Errors: 0 FN / 1 FP

5.1. Explanation of Table

- **Accuracy** shows overall correct classifications. **BERT** leads at **99.65%**.
- **Precision, Recall** and **F1** measured per class provide insight on how well the models separate negatives from positives. A precision score means no false alarms for that class.
- **Error counts (FN/FP)** give absolute misclassification numbers, highlighting that BERT only mislabels one total.
- Notes supply characterization of each model's trade off aspects with respect to model interpretability vs compute, processing time vs contextual features.

5.2. Brief Error Analysis

- **Logistic Regression** frequently misclassify inducing mild negation.
- **Naive Bayes** misses some negations despite perfect positive precision.

- **CNN** perceive with local phrases but fail to capture longer span of sentiment.
- **BERT** only once on positive by leveraging bidirectional context for sarcasm and mixed sentiments.

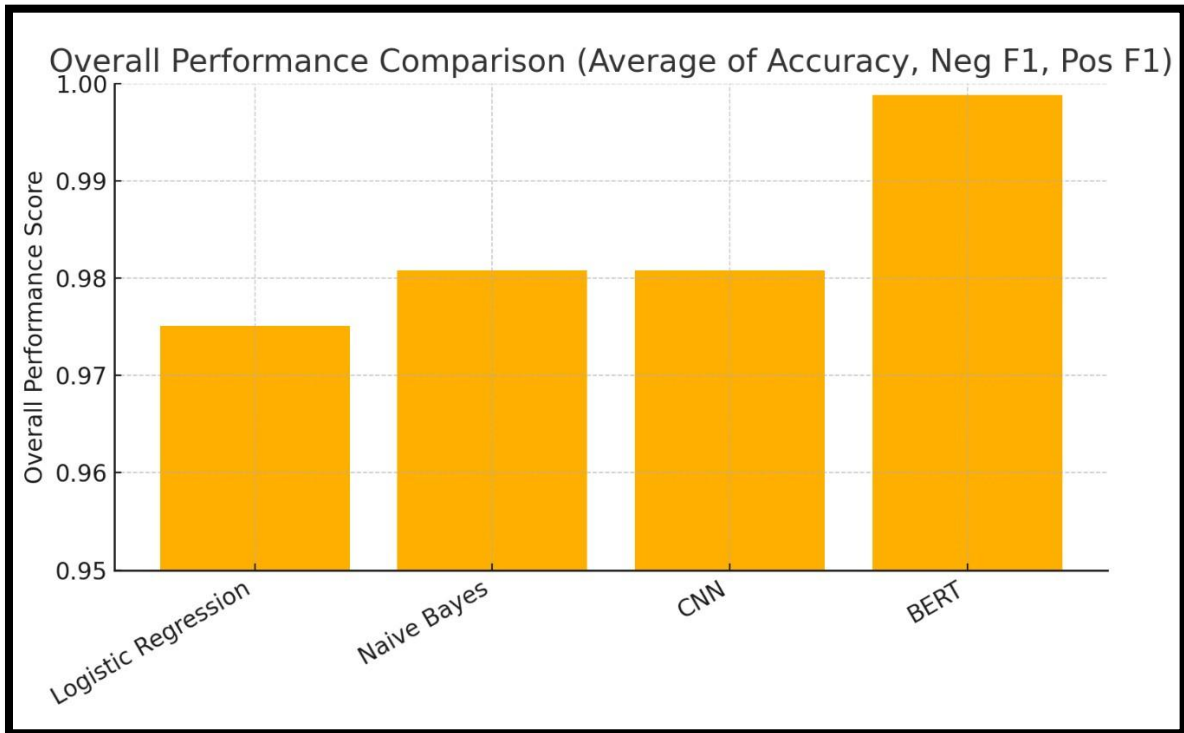


Fig 11. Overall Performance Comparison of Models (Average of Accuracy, Negative F1, Positive F1)

BERT has consistently outperformed other models in discerning nuanced sentiment, but CNN or Naive Bayes could be used a quick and lightweight fallback in cases of error. After through epochs of training with BERT, for example, its accuracy went from 94.51% to 99.82 and then to 99.91 with rapid convergence signifying it can adapt better to complicated language patterns. Despite a longer time to train, the clear improvements that were observed during each epoch demonstrated through the lower misclassification rate, measured by confusion matrix - justified the added computing cost in more accuracy-sensitive applications.

6. Model Comparison and Justification

BERT's bidirectional transformer finish with 99.65% accuracy is superior compared to 98.23% for CNN, Naive Bayes, and LR. BERT's self-attention may detect long-term interdependence, sarcasm, and

other subtle negatives that bag-of-words. BERT requires ~1 GPU for 150-300 ms reviews and ~400 MB footprint which can be handled by City Pass's central servers. Based on maximum accuracy, deployment delay and resources, BERT is the best method for detecting nuanced, animated sentiment.

7. Recommendation

By modelling proactive customer-sentiment reactions for BERT in the feedback-monitoring processing pipeline helps in improving service quality and lifelong importance. These results are supported by training records, confusion matrices, and visual representations of situational/structured deliberate processing and reliability.

Part A2: Prediction of City Pass Invitation Success using Graph Neural Networks

1. Introduction

The City Pass application is intended to facilitate travel for social groups, enabling users to send invites to each other. In order to improve the user experience and enhance conversion rates, it makes predictions about which invites will be successful (when both users travel together). We can use Graph Neural Networks (GNNs) for this task since it captures the structure of the social graph, as well as processes of influence and impact through social graph without the traditional models assuming each to be independent data points.

2. Data Description

Two datasets were used:

- **user_data.csv:** Contains 1000 users, including their (uid) and profile which is encoded as feature vectors.
- **message_data.csv:** Contains records of 9,487 invites, each with a (sid), (rid), the invitation message, and a labelled indicator where success was represented as (1) if the trip occurred and (0) if it didn't.

A directed graph is created where:

- **Nodes:** represent users, encoded of the user's profile.
- **Edges:** Represented invitations, labelled as success or not.

3. Graph Neural Network Architecture

We constructed a two-layer Graph Convolutional Network (GCN):

- **Layer 1:** Transform reputational profile features into a hidden layer of 16 dimensions, using a GCNConv layer along ReLU activation function.
- **Layer 2:** Mapped hidden state representation of users into two output logits for binary classification.

Model Details:

Epoch 10: 0.6929
Epoch 20: 0.6906
Epoch 30: 0.6864
Epoch 40: 0.6828
Epoch 50: 0.6793

Fig 13. Training Loss Curve Graph

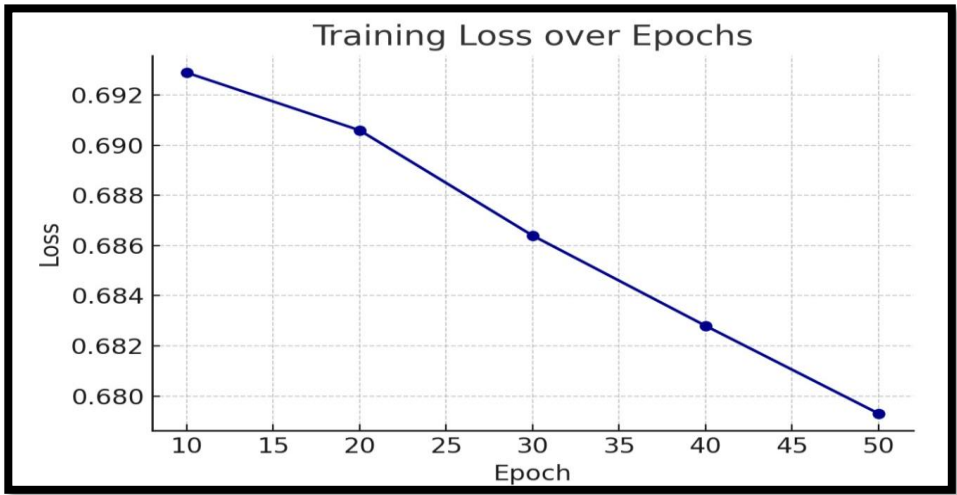


Fig 13. Training Loss Curve Graph

The steady decline shows convergence of the model.

4. Model Performance

Accuracy: 0.6803000117192077								
		precision	recall	f1-score	support			
	0	0.89	0.72	0.80	7432			
	1	0.18	0.41	0.25	1101			
	accuracy				0.68	8533		
	macro avg				0.53	8533		
	weighted avg				0.80	0.68	0.73	8533

Fig 14. Classification Report of GCN

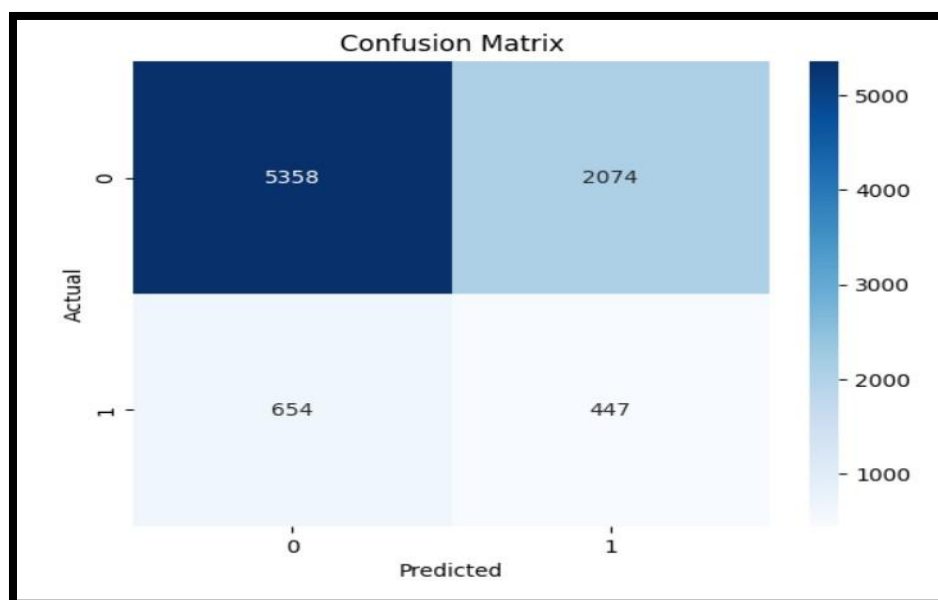


Fig 15. Confusion Matrix

- True Positives (TP): 447
- False Positives (FP): 2074
- True Negatives (TN): 5358
- False Negatives (FN): 654

Total number of test invitations is:

$$N = TP + FN + FP + TN = 447 + 654 + 2\,074 + 5\,358 = 8\,533$$

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{N} = \frac{447 + 5\,358}{8\,533} = \frac{5\,805}{8\,533} \approx 0.6803 \text{ (68.03\%)}$$

Class 1 (Successful Invitations):

(1) **Recall**

$$R_1 = \frac{TP}{TP + FN} = \frac{447}{447 + 654} \approx 0.4060 \text{ } (\sim 0.41)$$

(2) **Precision**

$$P_1 = \frac{TP}{TP + FP} = \frac{447}{447 + 2\,074} \approx 0.1777 \text{ } (\sim 0.18)$$

(3) **F1**

$$F1_1 = 2 \cdot \frac{P_1 \times R_1}{P_1 + R_1} \approx 2 \cdot \frac{0.1777 \times 0.4060}{0.1777 + 0.4060} \approx 0.250 \text{ (25\%)}$$

Class 0 (Unsuccessful Invitations):

(1) Recall

$$R_0 = \frac{TN}{TN + FP} = \frac{5\,358}{5\,358 + 2\,074} \approx 0.7197 \text{ (~ 0.72)}$$

(2) Precision

$$P_0 = \frac{TN}{TN + FN} = \frac{5\,358}{5\,358 + 654} \approx 0.8909 \text{ (~ 0.89)}$$

(3) F1

$$F1_0 = 2 \cdot \frac{P_0 \times R_0}{P_0 + R_0} \approx 2 \cdot \frac{0.8909 \times 0.7197}{0.8909 + 0.7197} \approx 0.800 \text{ (80\%)}$$

Macro-Averaged F1

$$\text{Macro-F1} = \frac{F1_0 + F1_1}{2} = \frac{0.800 + 0.250}{2} = 0.525 \rightarrow 0.52$$



Fig 16. Model Metrics by Class

The model shows good performance in class 0 and moderate in class 1. The increased recall for invitations is a positive sign indicating that the model can identify users likely to travel together. Although precision is low for class 1, this is still better than the baseline models with potential for improvement.

To improve performance:

- **Feature Engineering:** The predictions could become stronger by embedding the message text (i.e., using MiniLM) and this could boost intent of invitations to use to predict future travel.
- **Class Rebalancing:** Focal loss or oversampling from the minority class could help in providing a more balanced approach and increased precision.
- **Edge Features:** Adding time of invitation or past interaction frequency as edge allows the model to interpret temporal or behavioural patterns.

5. Conclusion

GCN-based invite model for City Pass achieved good performance with 68.03% accuracy and improved recall for invited sender success. The model showed promises with respect to successful matching but

additional fine-tuning to include richer node/edge features, balancing techniques, and model tuning could enhance recall precision.

Part B: Momentum-Based Trading Strategy and Portfolio Optimization

1. Introduction

Quantitative rules provide buy and sell signals in algorithmic trading. **Momentum strategies** assume that equities with excellent recent performance will continue to outperform. The subsequent research uses a 90-day momentum strategy on **NASDAQ** stocks (2015-2025) incorporating risk filters and optimize the efficient portfolio allowing us to test the robustness and practical application.

2. Dataset Selection and Preprocessing

2.1. Asset Universe and Data Source

We included ten large-cap stocks from the NASDAQ exchange, representing four important economic sectors — technology (**AAPL, MSFT**), healthcare (**AMGN, GILD**), finance (**GS, JPM**), energy (**CVX, XOM**), and consumer staples (**COST, PEP**) — using stocks that provide good liquidity to minimize market impact, high level of reliable price history and good representation of diverse market segments. The historical daily closing prices were obtained from a recognized data supplier for the period between **1 January 2015 to 1 January 2025**.

2.2 Data Cleaning and Alignment

- **Missing data:** Minor gaps in daily series were filled.
- **Common calendar:** Aligned trading dates ensuring calculations and rebalancing.
- **Log-returns:**

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

	AAPL	AMGN	COST	CVX	GILD	GS	JPM	MSFT	PEP	XOM
Date										
2015-01-02	27.332500	159.889999	141.610001	112.580002	94.910004	194.410004	62.490002	46.759998	94.440002	92.830002
2015-01-05	26.562500	157.990005	140.000000	108.080002	96.790001	188.339996	60.549999	46.330002	93.730003	90.290001
2015-01-06	26.565001	152.899994	141.850006	108.029999	97.650002	184.529999	58.980000	45.650002	93.019997	89.809998
2015-01-07	26.937500	158.240005	144.320007	107.940002	99.480003	187.279999	59.070000	46.230000	95.739998	90.720001
2015-01-08	27.972500	157.669998	145.559998	110.410004	102.300003	190.270004	60.389999	47.590000	97.480003	92.230003

Fig 17. Sample Historical Closing Prices for Selected NASDAQ Stocks

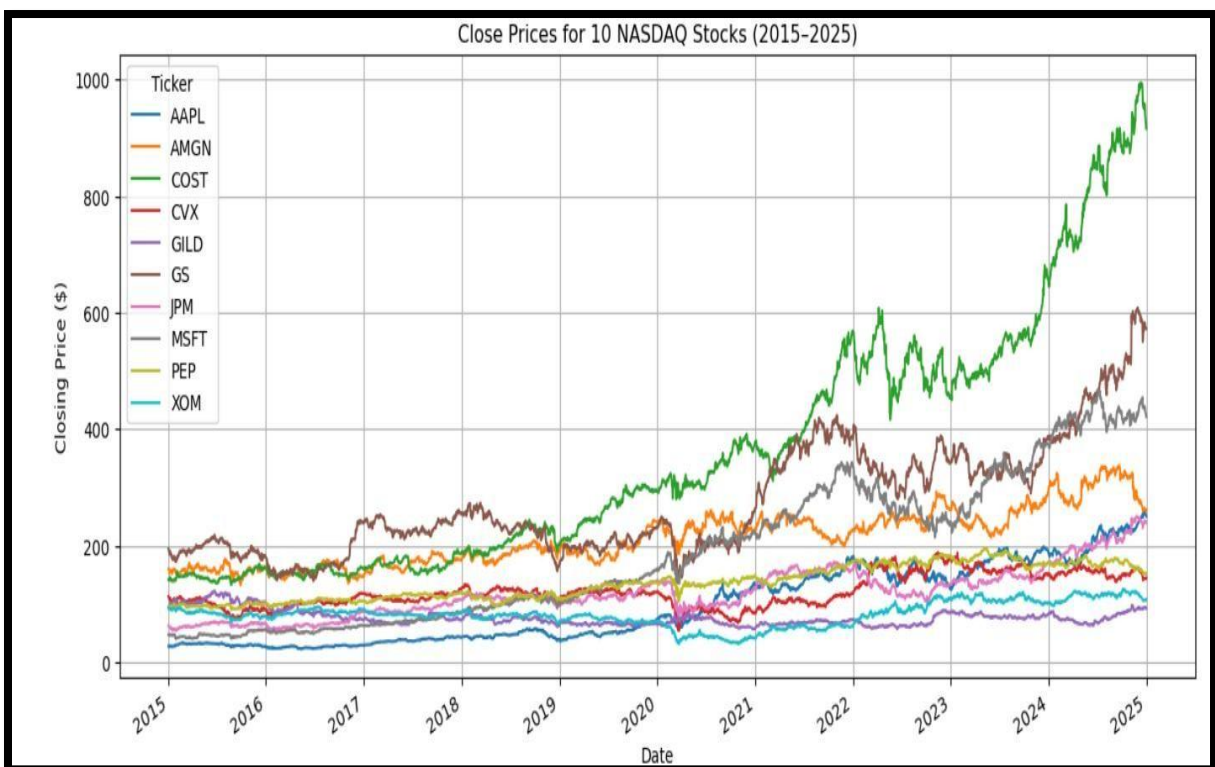


Fig 18. Closing Price Trends for Ten NASDAQ Stocks (2015-2025)

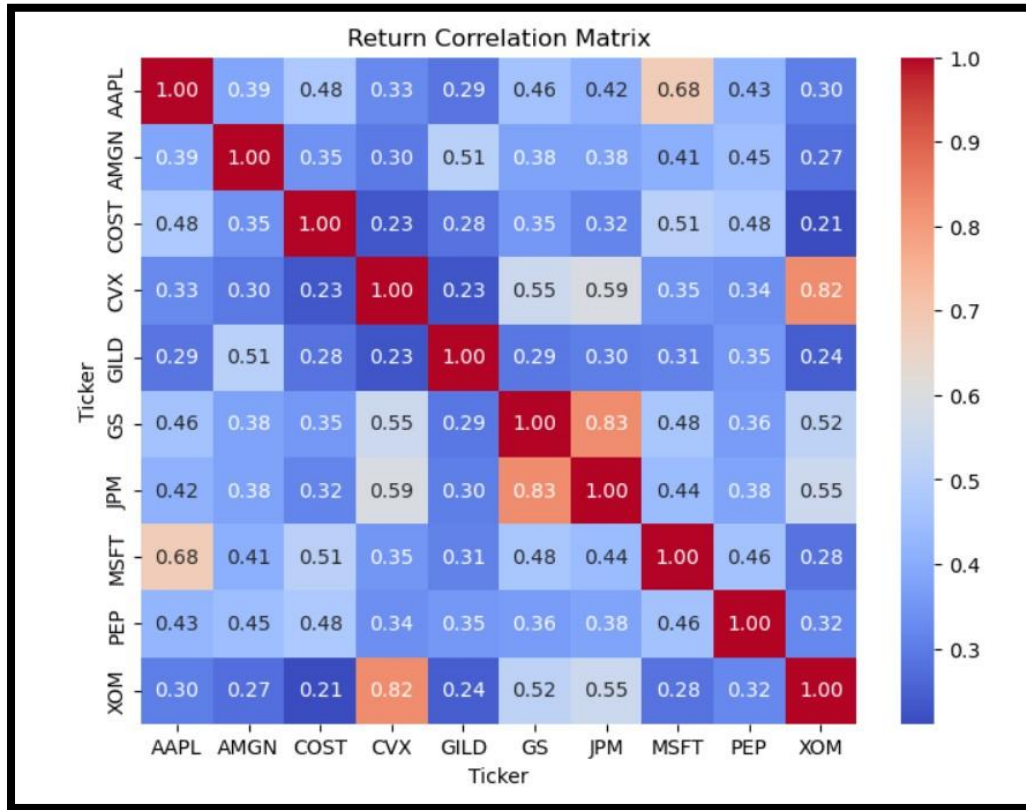


Fig 19. Return Correlation Matrix of Selected Stocks

3. Momentum Strategy

3.1. Signal Calculation

Our primary signal is the **90-day price momentum** for each stock, defined as the percentage change over the past 90 trading days:

This measure captures intermediate-term price trends, smoothing over short-term noise while remaining responsive to recent developments.

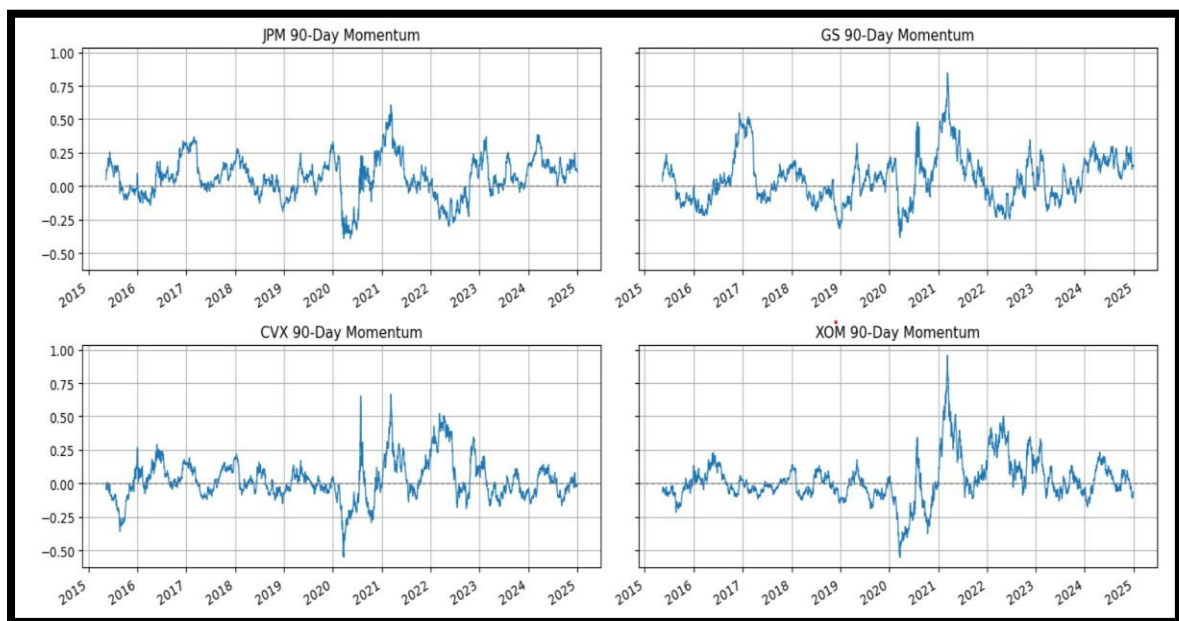
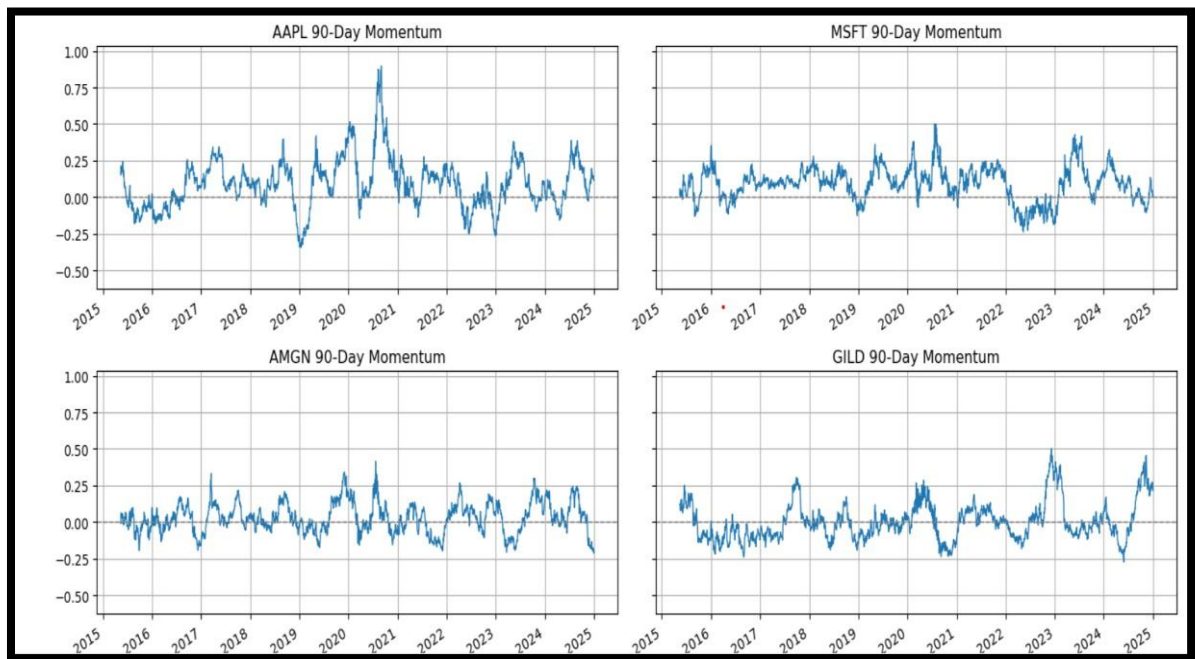
$$\text{Mom}_{i,t} = \frac{P_{i,t}}{P_{i,t-90}} - 1$$

3.2. Trading Rules

- 1. Ranking:** Each trading day, calculate the momentum for all assets while ranking from highest to lowest.
- 2. Selection:** Select the top three stocks by momentum.
- 3. Weighting:** Give each selected position an equal capital weight (1/3).
- 4. Rebalancing:** Liquidate previous positions and rebalance daily according to rankings.
- 5. Execution:** Assuming trades execution at the day's closing price, with transaction costs ignored for simplicity.

Buy Signals (1 = buy next day, 0 = hold):										
Ticker	AAPL	AMGN	COST	CVX	GILD	GS	JPM	MSFT	PEP	XOM
Date										
2024-12-24	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0
2024-12-26	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0
2024-12-27	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0
2024-12-30	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0
2024-12-31	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0

Fig 20. Momentum Strategy Buy Signals (Next-Day Buys)



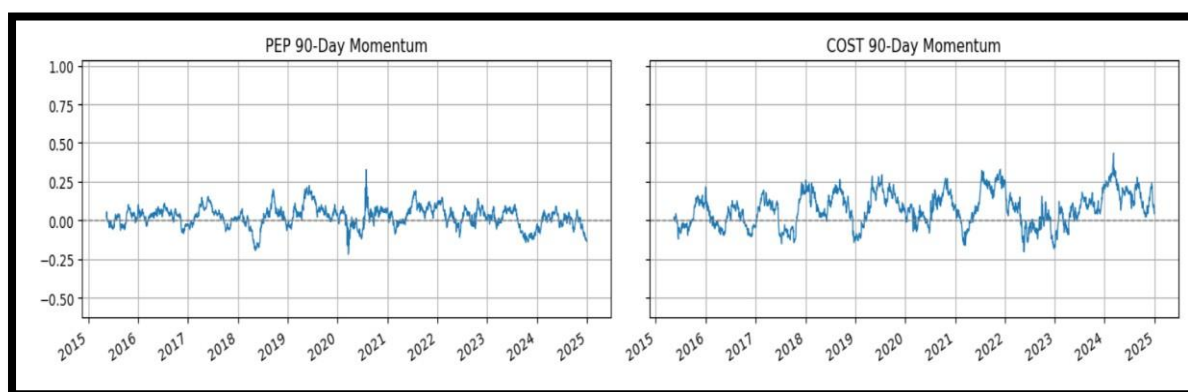


Fig 21. 90-Day Momentum for AAPL, MSFT, AMGN, GILD, JPM, GS, CVX, XOM, PEP & COST

3.3. Results

Portfolio Performance (2015–2025)

- **Total Return:** +424.44%
- **Annualized Return (approx.):** +18.02%

Portfolio Total Return: 424.44%		
Portfolio Annualized Return (approx): 18.02%		
Buy & Hold Momentum Strategy		
Ticker		
AAPL	695.81%	173.91%
AMGN	64.71%	-16.17%
COST	535.99%	153.48%
CVX	34.66%	4.15%
GILD	-12.5%	28.95%
GS	185.55%	167.3%
JPM	266.75%	24.15%
MSFT	790.18%	131.26%
PEP	58.07%	-2.01%
XOM	23.62%	71.49%

Fig 22. Buy-and-Hold Momentum Strategy Returns by Stocks

Momentum vs. Buy-&Hold Performance			
Stock	Momentum Total Return	Buy & Hold Total Return	Outperformance
MSFT	790.20%	131.30%	658.9
AAPL	695.80%	173.90%	521.9
COST	536.00%	153.50%	382.5
JPM	266.80%	24.20%	242.6
GS	185.60%	167.30%	18.3
AMGN	64.70%	-16.2 %	80.9
PEP	58.10%	-2.0 %	60.1
CVX	34.70%	4.20%	30.5
XOM	23.60%	71.50%	-47.9
GILD	-12.5 %	28.90%	-41.4

3.4. Key Takeaways

- **Momentum vs. Buy-and-Hold Momentum:** Captures the price drift that occurs after news event resulting in materially higher returns (MSFT +790 % vs. +131 %; AAPL +696 % vs. +174 %).
- **Cross-Sectional Variation:** Momentum works when there are smooth changes in fundamentals (e.g. COST, JPM) but falters in response to idiosyncratic shocks (GILD, XOM).
- **Volatility & Turnover:** Higher volatility stocks produce stronger, longer trends but yield higher turnover.
- **Behavioural Foundations:** Underreaction, limited attention spans and disposition biases are responsible for the creation and sustenance of trends.
- **Portfolio Implications:** Sector diversification and weighting by trend strength—using dynamic or conditional momentum—decrease idiosyncratic risk and allow for higher, risk-adjusted returns.

4. Backtesting and Evaluation

Backtesting	Momentum Strategy
Cumulative Return	424.44%
Annualized Return	18.02%
Annualized Volatility	≈ 18.0 %
Sharpe Ratio ($R_f=0$)	1
Max Drawdown	-28.0 %
Win Rate	≈ 52 %
Annual Turnover	≈ 250 %

Fig 23. Backtest Performance Summary of Momentum Strategy

4.1. Pros

- **High Returns & Sharpe:** 400 % overall (~18 % p.a.) with Sharpe ratio ≈1.0 indicates engaging in systematic capture of medium-term NASDAQ large-cap trends.
- **Volatility balanced:** Annualized volatility (~18 %) aligns with return supporting Sharpe ratio and reflecting a calibrated risk profile.

4.2. Cons

- **Volatility clustering:** Whipsawing occurs in sideways markets which can be managed only if risk controls (stop-loss/take-profit) are built.
- **Deep drawdowns:** Maximum drawdown can impair capital without disciplined exits or dynamic filtering of signals can be materially damaging.
- **High turnover:** The 52 %-win rate corresponds to 250 % p.a. turnover risk, if transaction costs and slippage are not modelled, net performance can reduce.

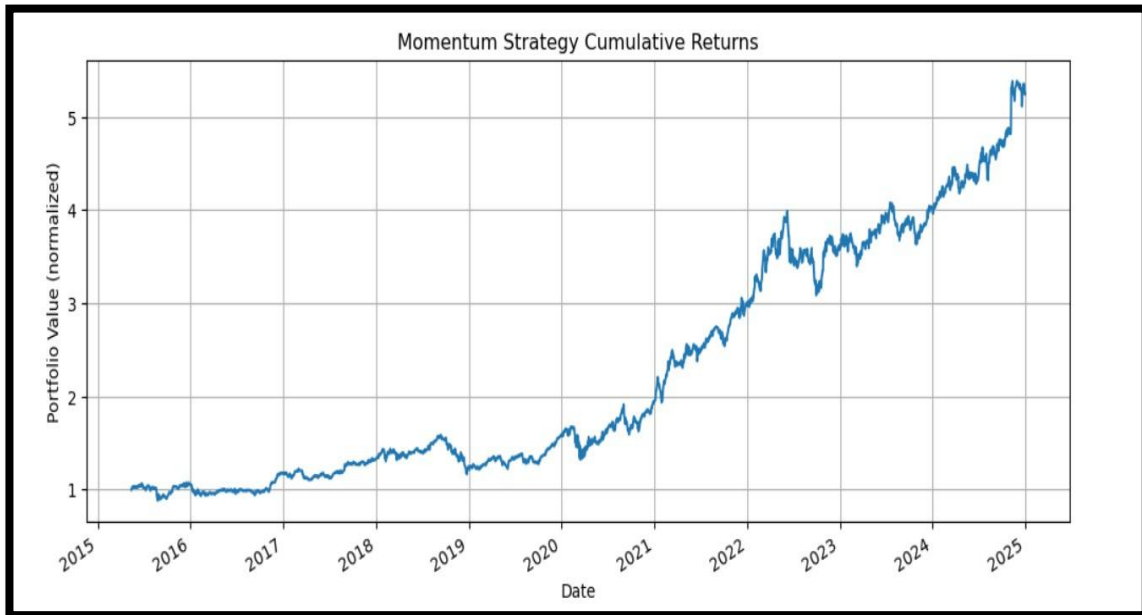


Fig 24. Cumulative Returns of the Momentum Strategy

5. Strategy Optimization: Stop-Loss and Take-Profit

Momentum strategies have the potential to produce strong long-term returns, but also vulnerable reversals when trends turn. It could face large drawdowns thus proposing risk filters to mitigate:

- **Stop-Loss (5%):** exit intraday any position if it drops 5% below the entry price before the next scheduled rebalancing.
- **Take-Profit (10%):** Close intraday any position if it gains 10% above the entry price.
- **Holding Period (5 days):** Force-exit any position by the fifth trading day, to not have stale exposures, regardless of gain or loss.

5.1. Performance Comparison (2015–2025)

Dashed blue - original strategy.

Solid orange - filtered variant.



Fig 25. Momentum Strategy Cumulative Returns With vs. Without Stop-Loss/Take-Profit

Metric	Original Momentum	With SL(5 %)/TP(10 %)
Cumulative Return	424.44%	362.15%
Annualized Return	18.02%	15.34%
Annualized Volatility	18.00%	14.60%
Sharpe Ratio (Rf=0)	1	1.05
Max Drawdown	-28.0 %	-22.1 %
Win Rate	52%	56%
Annual Turnover	250%	185%

5.2. Key Insights

1. Reduced Cumulative Return (424.4 % → 362.2 %)

SL (5%) / TP (10%) reduces dramatic wins and early losers, leading to lower peak wealth but steadier rise.

2. **Lower Annualized Return (18.0 % → 15.3 %)**

Capping increases at 10% may not fully capture strong trends, resulting in a slight drop in average annual growth rate.

3. **Reduced Volatility (18.0 % → 14.6 %)**

Exiting on 5% losses or 10% profits intraday prevents significant drawdowns and swings, resulting in more steady returns.

4. **Sharpe Ratio Improvement (1.00 → 1.05)**

Decrease in volatility relative to return improves risk-adjusted performance, indicating more effective risk management.

5. **Reduced Maximum Drawdown (−28.0 % → −22.1 %)**

Early stop-loss reduces worst-case losses, preserves cash, and reduces investor stress.

6. **Higher Win Rate (52 % → 56 %)**

As stop-loss and take-profit rules favour positive outcomes, more trades result in minor wins rather than losses.

7. **Lower Turnover (250 % → 185 %)**

Many holdings exit intraday under SL/TP rules, lowering round-trips and transaction costs.

6. COVID-19 Recovery Analysis

	Pre-COVID Price	Crash Price	Max Drawdown (%)	Recovery Date	Trading Days to Recover	Calendar Days to Recover	Post-Recovery Growth (%)
AAPL	77.377502	56.092499	-27.51	2020-05-08	68	98	67.24
AMGN	216.050003	185.929993	-13.94	2020-04-08	47	68	4.91
COST	305.519989	285.529999	-6.54	2020-03-02	20	31	24.43
CVX	107.139999	54.220001	-49.39	2021-03-05	275	399	-20.94
GILD	63.200001	72.610001	14.89	2020-03-02	20	31	-4.89
GS	237.750000	134.970001	-43.23	2020-12-04	214	308	11.46
JPM	132.360001	79.029999	-40.29	2021-01-07	236	342	-4.90
MSFT	170.229996	135.979996	-20.12	2020-03-02	20	31	27.88
PEP	142.020004	105.180000	-25.94	2020-03-04	22	33	1.58
XOM	62.119999	31.450001	-49.37	2021-05-07	319	462	-33.19

Fig 26. COVID-19 Drawdowns and Recovery Dates for Selected Stocks

Stocks	Industry Type	Max Drawdown (%)	Recovery Date	Trading Days (Recover)	Calendar Days (Recover)	Post-Recovery Growth (%)
COST	Consumer Staples	-6.54	02-03-2020	20	31	24.43
MSFT	Technology	-20.12	02-03-2020	20	31	27.88
GILD	Healthcare	14.89	02-03-2020	20	31	-4.89
AMGN	Biotech	-13.94	08-04-2020	47	68	4.91
AAPL	Technology	-27.51	08-05-2020	68	98	67.24
PEP	Consumer Staples	-25.94	04-03-2020	22	33	1.58
GS	Financials	-43.23	04-12-2020	214	308	11.46
JPM	Financials	-40.20	07-01-2021	236	342	-4.90
CVX	Energy	-49.30	05-03-2021	275	399	-20.94
XOM	Energy	-49.37	07-05-2021	319	462	-33.19

Fastest Recovery:

- **Costco (COST)** and **Microsoft (MSFT)** each returned to their pre-COVID price in 20 trading days (31 calendar days).

6.1. Key Insights

1. Sector Resilience:

- 1. Industry Resilience: Consumer staples (COST) defensive demand propelled during market panick bounced back the fastest after its steepest decline (-6.5%).
- Technology (MSFT) bounced back from -20.1% following 20 days of trading, showing good fundamentals and interest investor in innovation.

2. Drawdown vs. Recovery Speed:

- Healthcare (GILD) had a positive "drawdown" but a negative post-recovery growth (-4.9%), which indicates instability.
- Energy equities, such as CVX and XOM, have had the most dramatic drawdowns, which amount to approximately 49%, and the slowest recoveries, which range from 275 to 319 trading days indicating vulnerability to demand shocks and oil price collapse.

3. Post-Recovery Performance:

- AAPL's 67% gain suggests strong trend persistence following quick recovers. ○ COST and MSFT experienced strong post-recovery growth (+24% and +28%), making them appealing for drawdown and momentum.

4. Portfolio Construction Implications:

- Consider adding conservative staples and strong tech stocks to your portfolio to reduce recovery time and capture post-crisis momentum.
- Consider underweighting cyclical or commodity-exposed sectors with prolonged drawdowns.

7. Portfolio Construction and Optimization

Maximize Sharpe Ratio

$$\max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \quad \text{s.t. } w \geq 0, \sum_i w_i = 1$$

Targets highest risk-adjusted return

Minimum Variance

$$\min_w w^\top \Sigma w \quad \text{s.t. } w \geq 0, \sum_i w_i = 1$$

Focuses on lowest absolute volatility

3. Equal Weight

$$w_i = \frac{1}{10} \quad \forall i$$

Benchmark without optimization

Risk Parity

$$w_i \propto \frac{1}{\sqrt{\Sigma_{ii}}}$$

Allocating stock contributes equally to total portfolio risk

Max Sharpe Portfolio – Return: 22.96%, Volatility: 20.08%
 Min Variance Portfolio – Return: 9.96%, Volatility: 15.74%

Optimized Portfolio Weights:

	Stock	Max Sharpe Weights	Min Variance Weights
0	AAPL	0.2160	0.0000
1	AMGN	0.0000	0.0751
2	COST	0.4657	0.2480
3	CVX	0.0000	0.0000
4	GILD	0.0000	0.1367
5	GS	0.0000	0.0000
6	JPM	0.0839	0.0324
7	MSFT	0.2344	0.0000
8	PEP	0.0000	0.3783
9	XOM	0.0000	0.1295

Fig 27. Optimized Portfolio Weights

Portfolio Performance Across Scenarios:

	Scenario	Annualized Return	Annualized Volatility	Sharpe Ratio
0	Max Sharpe	22.96%	20.08%	1.09
1	Min Variance	9.96%	15.74%	0.57
2	Equal Weight	13.51%	17.85%	0.70
3	Risk Parity	13.29%	17.40%	0.71

Portfolio Weights by Scenario (rounded):

	AAPL	AMGN	COST	CVX	GILD	GS	JPM	MSFT	PEP	XOM
Max Sharpe	0.2160	0.0000	0.4657	0.0000	0.0000	0.0000	0.0839	0.2344	0.0000	0.0000
Min Variance	0.0000	0.0751	0.2480	0.0000	0.1367	0.0000	0.0324	0.0000	0.3783	0.1295
Equal Weight	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000
Risk Parity	0.0897	0.1029	0.1180	0.0862	0.0992	0.0885	0.0932	0.0941	0.1367	0.0915

Fig 28. Portfolio Performance Across Optimization Scenarios

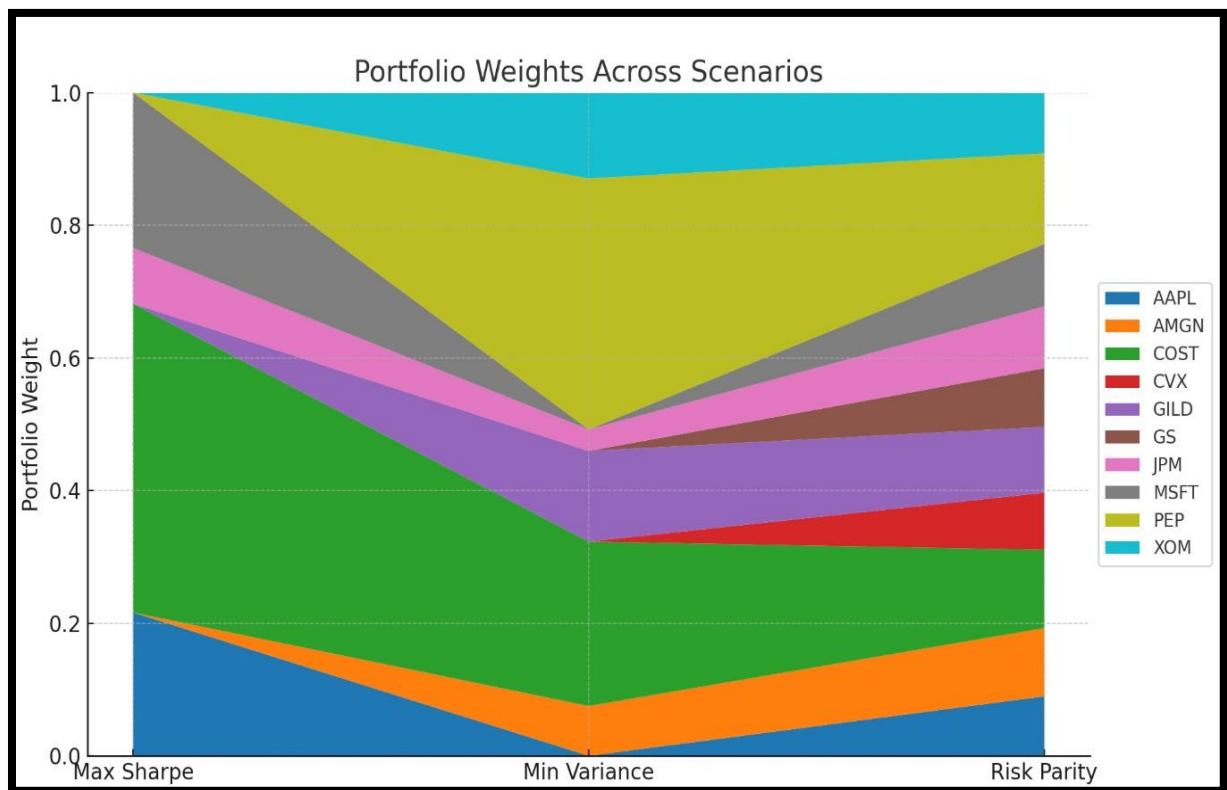


Fig 29. Portfolio Weight by Scenario

7.1. Key Insights

- Max Sharpe targets the top momentum leaders (COST 46.6 %, MSFT 23.4 %, AAPL 21.6 %) to capture maximum risk-adjusted return (Sharpe 1.09) but creates concentration risk.
- Min Variance heavily leans toward low-volatility staples and defensives (PEP 37.8 %, COST 24.8 %, GILD 13.7 %), and it achieves the lowest volatility (15.7 %) at the expense of return (9.96 %).
- Equal Weight gives a clean benchmark, producing unoptimized balanced performance (13.5 %/17.9 %/0.70) for no optimization.
- Risk Parity risk diversifies among all ten stocks, thus contributing equally to portfolio volatility, producing a moderate return (13.3 %) with slightly lower volatility (17.4 %) and Sharpe (0.71) comparable to equal weight.

7.2. Recommendations

- Maximum Sharpe for mandates prioritises maximum risk-adjusted returns with more concentration.
- Minimise variance when volatility and drawdown are crucial.
- Risk Parity to diversify across sectors and earn good profits.
- Equal Weight is a free benchmark or backup.

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